

3-29-assignment

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Meta-Analysis

```
library(metafor)
```

```
## Warning: package 'metafor' was built under R version 4.2.2
```

```
## Loading required package: Matrix
```

```
## Loading required package: metadat
```

```
## Warning: package 'metadat' was built under R version 4.2.2
```

```
##
```

```
## Loading the 'metafor' package (version 3.8-1). For an  
## introduction to the package please type: help(metafor)
```

```
adhd_data = read.csv('C:/Users/jacma/Downloads/ADHD.csv')  
yi = adhd_data[,16]  
vi = adhd_data[,17]  
analysis = rma(yi,vi,measure='RD')  
summary(analysis)
```

```
##
```

```
## Random-Effects Model (k = 41; tau2 estimator: REML)
```

```
##
```

```
##      logLik   deviance      AIC      BIC      AICc  
## -202.2401   404.4801   408.4801   411.8579   408.8045
```

```
##
```

```
## tau2 (estimated amount of total heterogeneity): 349.0573 (SE = 239.2865)
```

```
## tau (square root of estimated tau2 value):      18.6831
```

```
## I2 (total heterogeneity / total variability):    32.87%
```

```
## H2 (total variability / sampling variability):    1.49
```

```
##
```

```
## Test for Heterogeneity:
```

```
## Q(df = 40) = 63.4688, p-val = 0.0105
```

```
##
```

```
## Model Results:
```

```
##
## estimate      se      zval      pval      ci.lb      ci.ub
## 67.3725  5.4115  12.4499  <.0001  56.7662  77.9789  ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-Regression

```
mod1 = adhd_data[,12]
reg1 = rma(yi,vi,mods=~mod1)
```

```
## Warning: Studies with NAs omitted from model fitting.
```

```
summary(reg1)
```

```
##
## Mixed-Effects Model (k = 35; tau^2 estimator: REML)
##
##      logLik    deviance      AIC      BIC      AICc
## -166.7712   333.5423   339.5423   344.0319   340.3699
##
## tau^2 (estimated amount of residual heterogeneity):      218.9710 (SE = 230.0648)
## tau (square root of estimated tau^2 value):              14.7977
## I^2 (residual heterogeneity / unaccounted variability):  22.26%
## H^2 (unaccounted variability / sampling variability):     1.29
## R^2 (amount of heterogeneity accounted for):               31.12%
##
## Test for Residual Heterogeneity:
## QE(df = 33) = 44.8954, p-val = 0.0811
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 3.4841, p-val = 0.0620
##
## Model Results:
##
##      estimate      se      zval      pval      ci.lb      ci.ub
## intrcpt    16.8440  24.4404  0.6892  0.4907  -31.0583  64.7463
## mod1         0.0779   0.0417  1.8666  0.0620   -0.0039   0.1597 .
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```